ICG Presentation

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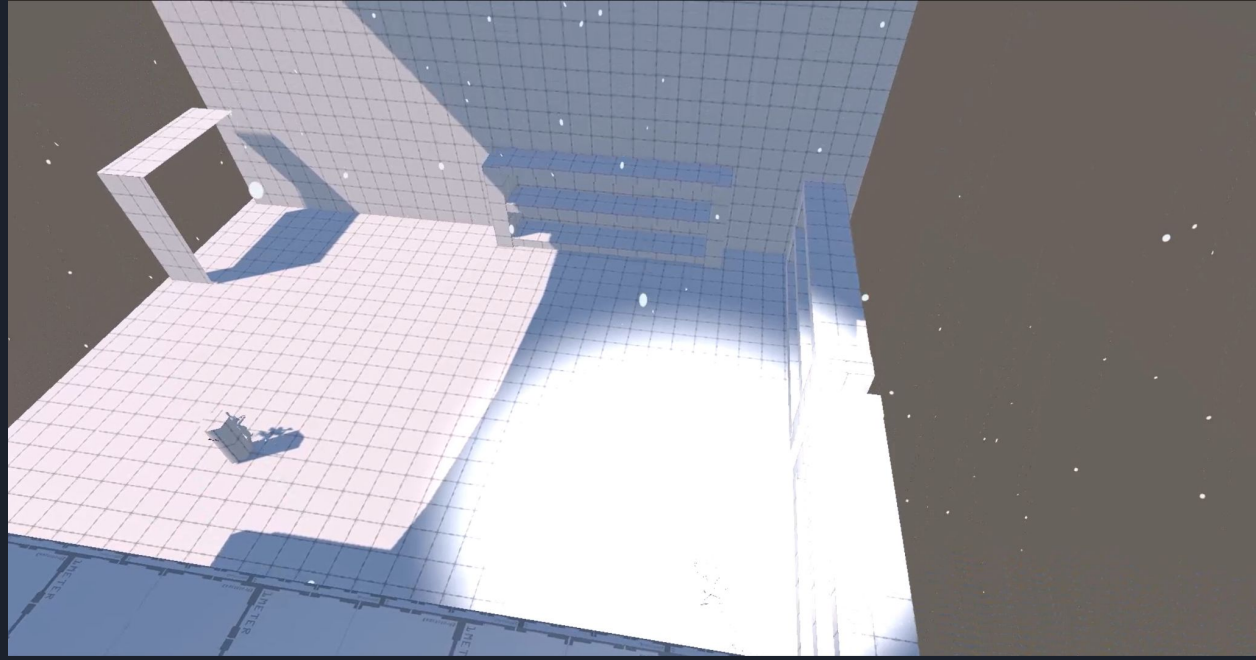
BATMAN
ARKHAM ORIGINS

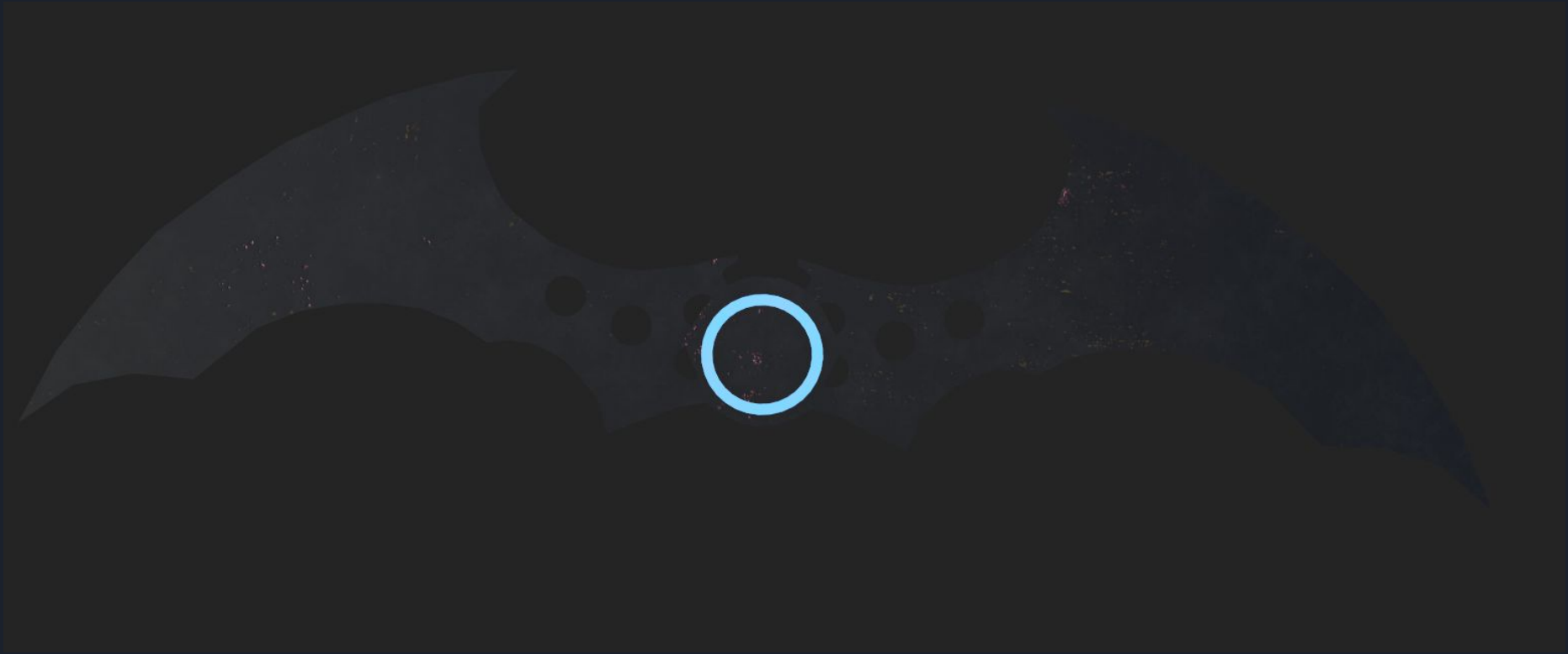
Centered around combat encounter in warehouse area inspired by promotional trailer

Predator fight where Batman systematically takes down ninjas



Mapped out simple layout for map level, and placed cameras in all the positions I would need them in for the cutscene.

Afterwards I looked for rigged models to place around in my scene.

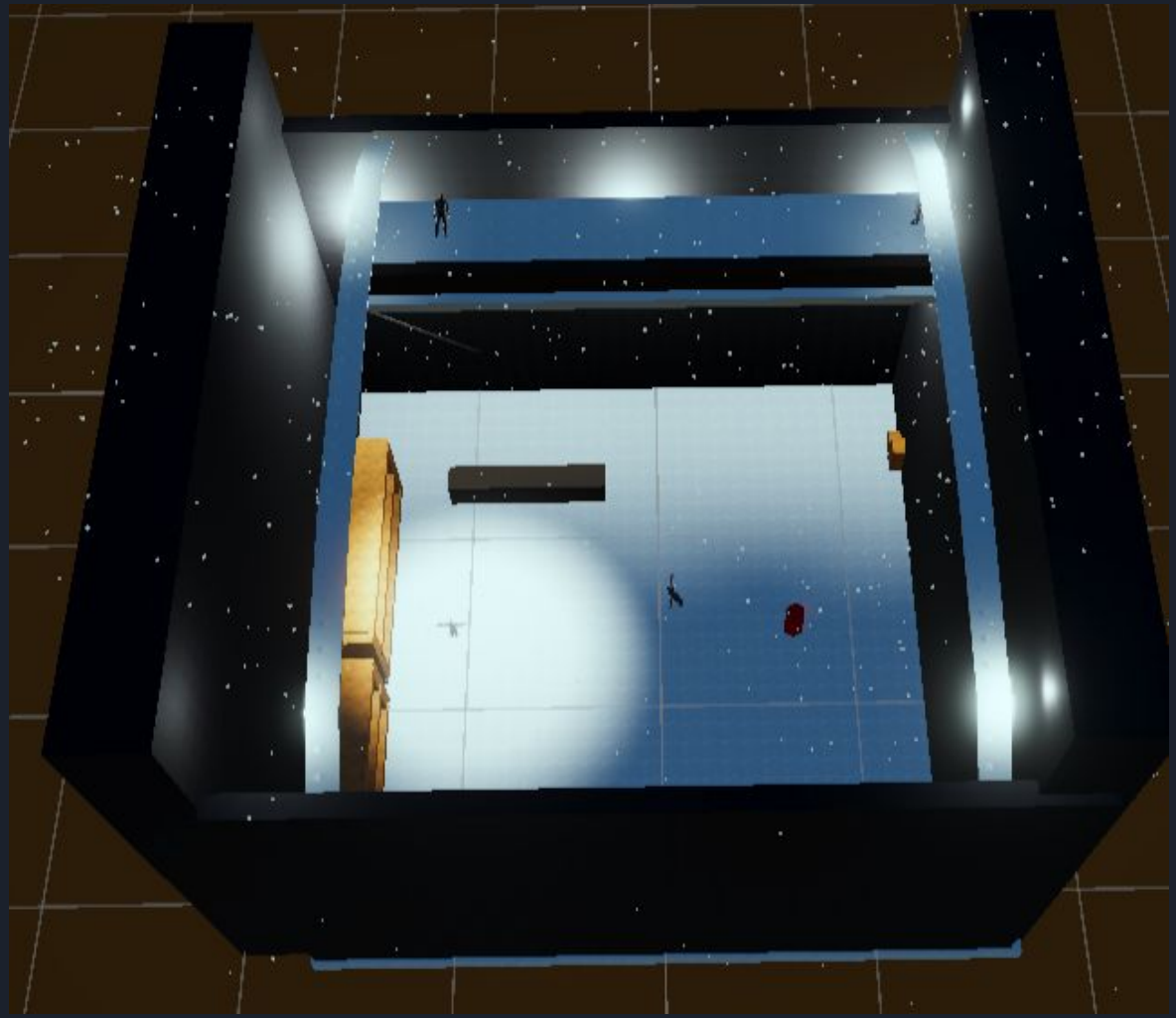


Found some models for elements of batman’s gadgets, but I didn’t find a good batarang, making me create my own model and texture for it.

Once I had the models, I began implementing animations that I got from mixamo to create choregraphy.

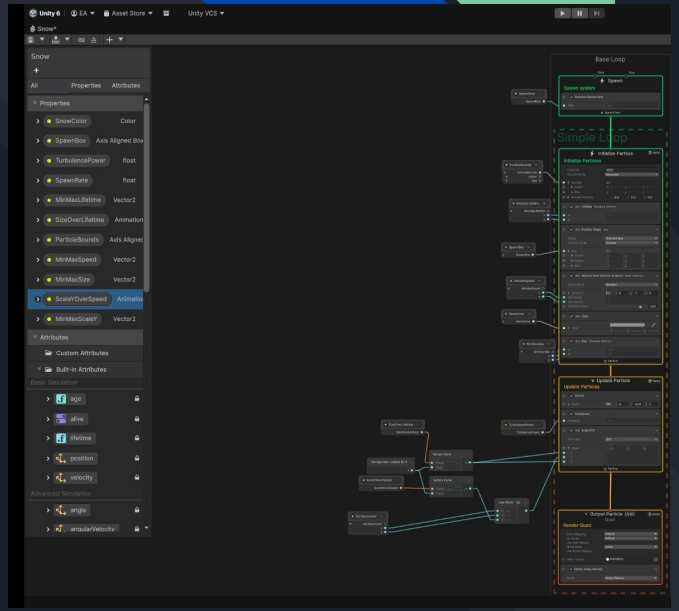
I added additional lights after populating the scene, since the models were too hard to see.

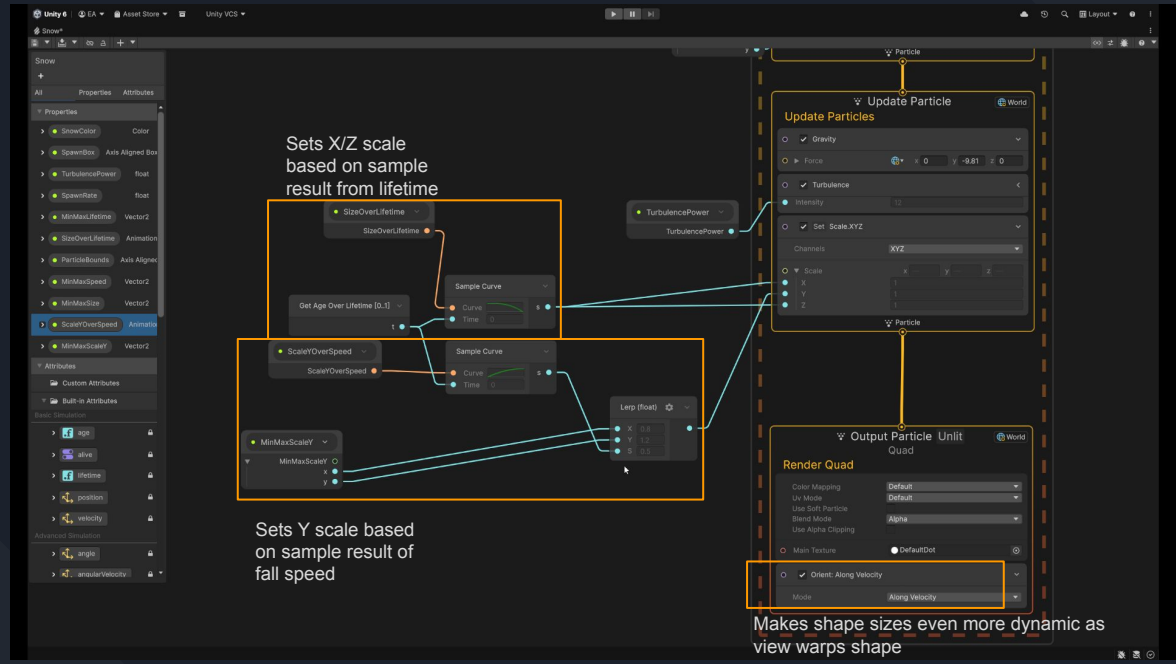
I also applied some textures to the terrain to make the environment more readable.



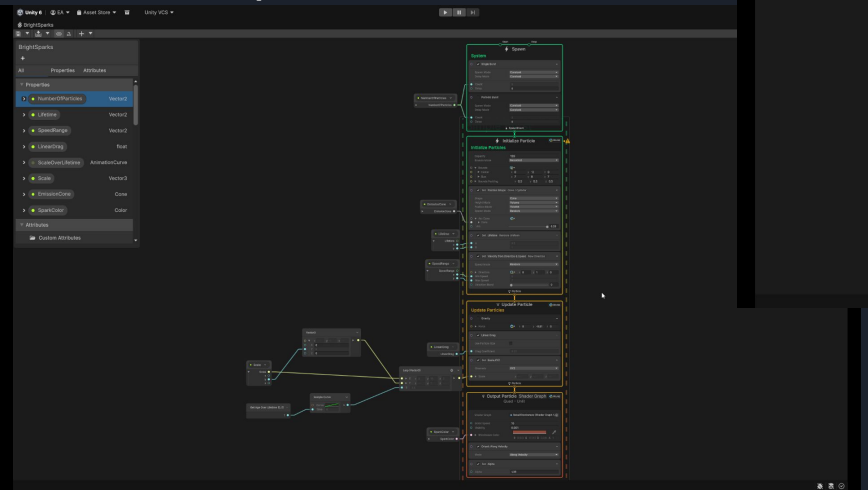
Once the scene was completed, I then went through the cutscene and implmented VFX for certain elements of the scene.

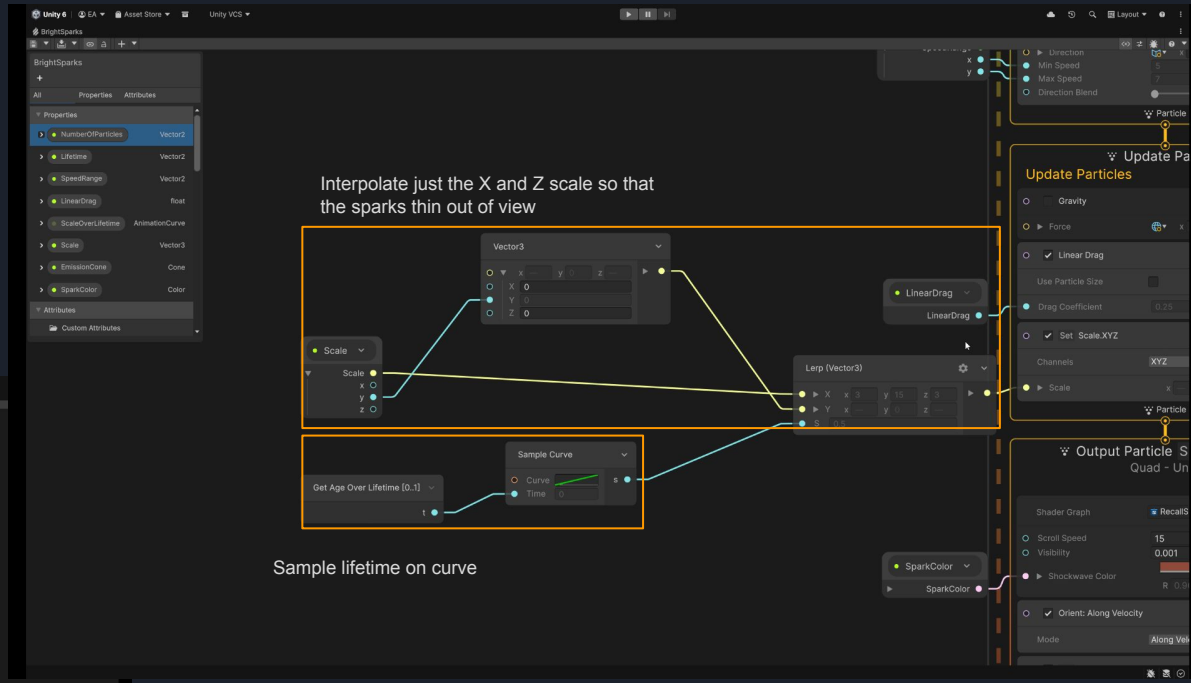
Full Graph

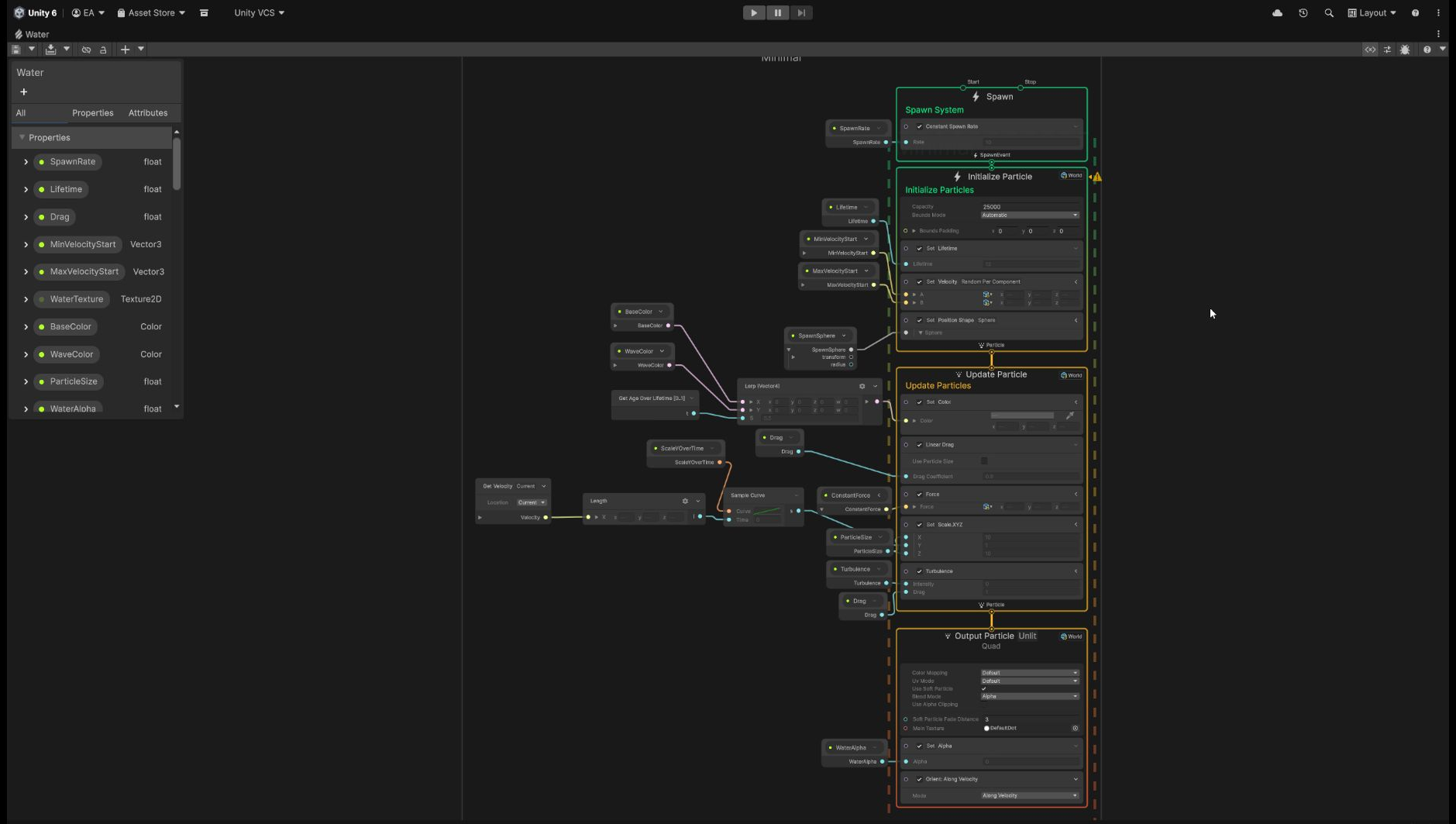


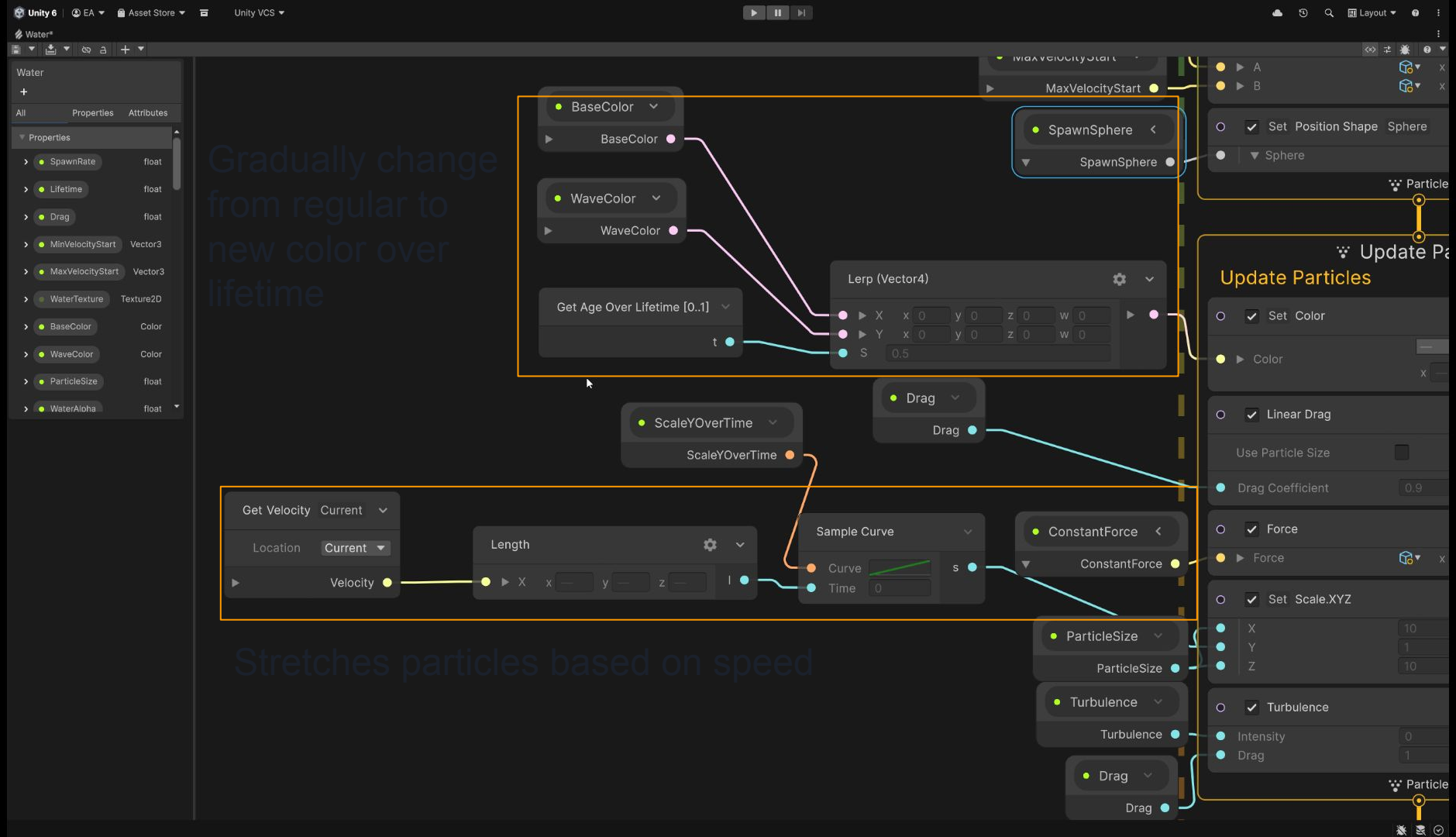

Sets X/Z scale based on sample result from lifetime
Sets Y scale based on sample result of fall speed
Makes shape sizes even more dynamic as view warps shape
Update Particle
Update Particles
Output Particle Unlit
Quad
Render Quad
Sample Curve
Lerp (float)
Get Age Over Lifetime [0..1]
SizeOverLifetime
ScaleYOverSpeed
MinMaxScaleY
TurbulencePower

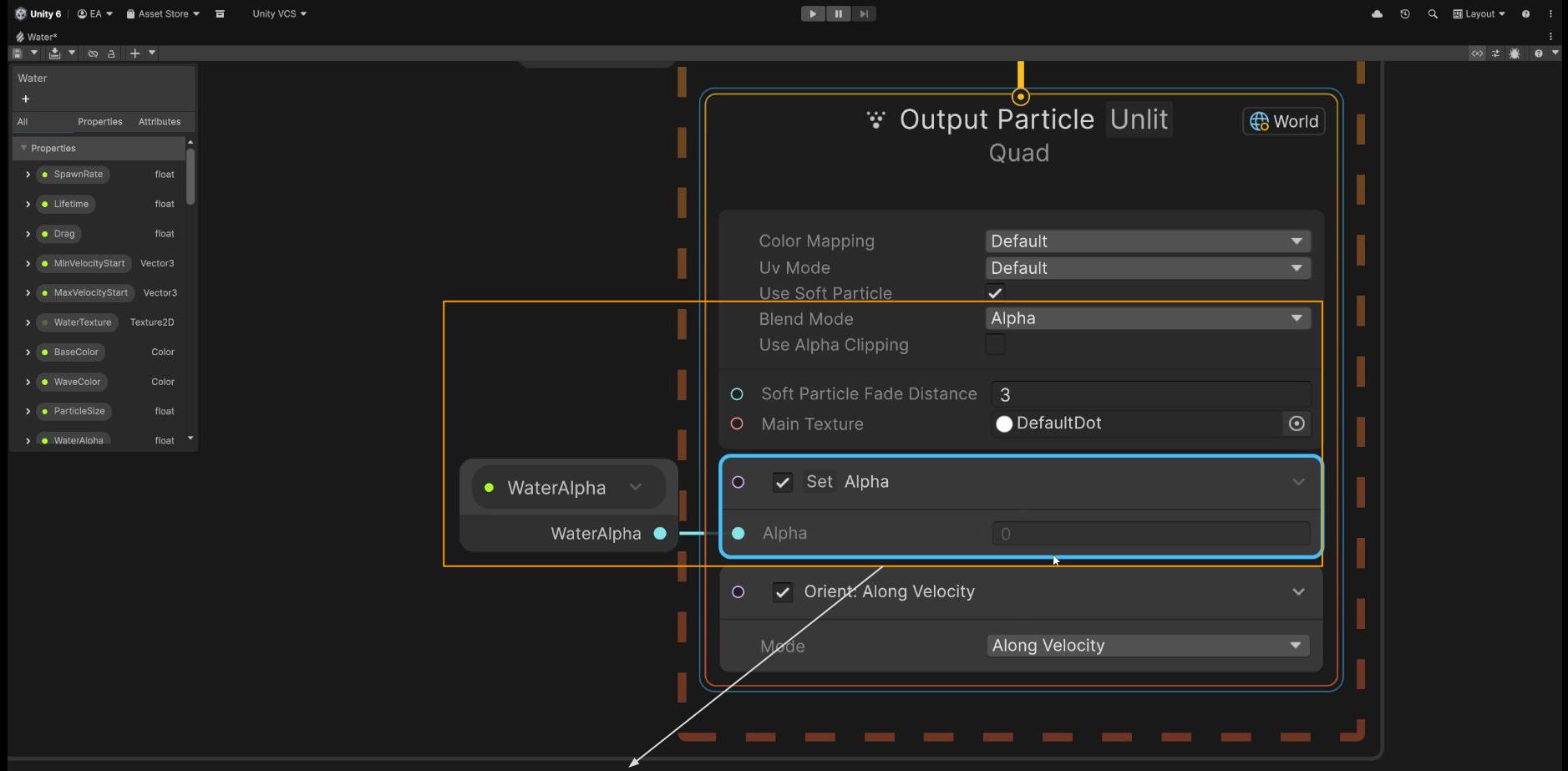
Full Graph



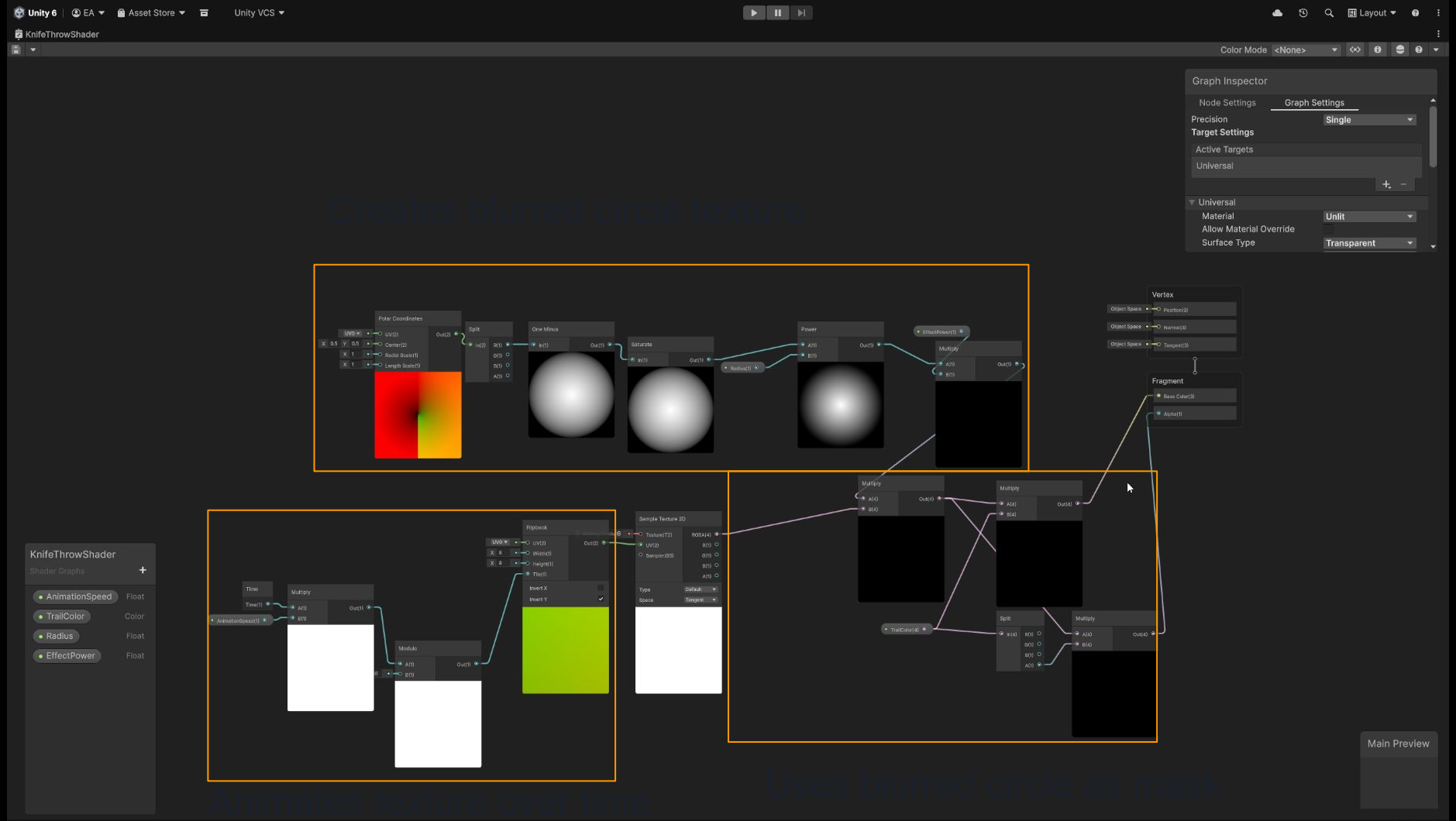

Interpolate just the X and Z scale so that the sparks thin out of view
Sample lifetime on curve
Vector3
Scale
Lerp (Vector3)
LinearDrag
Sample Curve
Get Age Over Lifetime [0..1]
SparkColor
Update Particles
Output Particle
BrightSparks
NumberOfParticles
Lifetime
SpeedRange
LinearDrag
ScaleOverLifetime
Scale
EmissionCone
SparkColor


Unity 6
EA
Asset Store
Unity VCS
Layout
Water
Water
All
Properties
Attributes
Properties
SpawnRate float
Lifetime float
Drag float
MinVelocityStart Vector3
MaxVelocityStart Vector3
WaterTexture Texture2D
BaseColor Color
WaveColor Color
ParticleSize float
WaterAlpha float
Minimal
Start
Stop
Spawn
Spawn System
Constant Spawn Rate
Rate
SpawnEvent
Initialize Particle
World
Initialize Particles
Capacity
25000
Bounds Mode
Automatic
Bounds Padding
Set Lifetime
Lifetime
Set Velocity Random Per Component
Set Position Shape Sphere
Sphere
Particle
Update Particle
Update Particles
Set Color
Color
Linear Drag
Use Particle Size
Drag Coefficient
Force
Set Scale.XYZ
Turbulence
Intensity
Drag
Output Particle Unlit
Quad
Color Mapping
Default
Uv Mode
Use Soft Particle
Blend Mode
Alpha
Use Alpha Clipping
Soft Particle Fade Distance
Main Texture
DefaultDot
Set Alpha
Alpha
Orient Along Velocity
Mode
Along Velocity
SpawnRate
Lifetime
MinVelocityStart
MaxVelocityStart
BaseColor
WaveColor
SpawnSphere
transform
radius
Lerp (Vector4)
Get Age Over Lifetime [0..1]
ScaleVOverTime
Drag
Get Velocity Current
Location
Velocity
Length
Sample Curve
Curve
Time
ConstantForce
ParticleSize
Turbulence
WaterAlpha


Unity 6
EA
Asset Store
Unity VCS
Layout
Water*
Water
All
Properties
Attributes
Properties
SpawnRate float
Lifetime float
Drag float
MinVelocityStart Vector3
MaxVelocityStart Vector3
WaterTexture Texture2D
BaseColor Color
WaveColor Color
ParticleSize float
WaterAlpha float
Gradually change from regular to new color over lifetime
BaseColor
BaseColor
WaveColor
WaveColor
Get Age Over Lifetime [0..1]
t
Lerp (Vector4)
X
Y
S
0.5
MaxVelocityStart
SpawnSphere
SpawnSphere
Set Position Shape Sphere
Sphere
Particle
Update Particles
Set Color
Color
Linear Drag
Use Particle Size
Drag Coefficient
0.9
Drag
Drag
ScaleYOverTime
ScaleYOverTime
Get Velocity Current
Location Current
Velocity
Length
Sample Curve
Curve
Time
s
ConstantForce
ConstantForce
Force
Force
Set Scale.XYZ
X
Y
Z
10
1
10
Stretches particles based on speed
ParticleSize
ParticleSize
Turbulence
Turbulence
Turbulence
Intensity
Drag
Drag
Drag
Particle



Creates the illusion of flame by overlapping particles, making objects near the center overlap more and thus have higher alpha, whereas further values have smaller alphas. The base alpha helps to blend particles together.


Unity 6
EA
Asset Store
Unity VCS
Layout
KnifeThrowShader
Color Mode
<None>
Graph Inspector
Node Settings
Graph Settings
Precision
Single
Target Settings
Active Targets
Universal
Universal
Material
Unlit
Allow Material Override
Surface Type
Transparent
Creates blurred circle texture
Polar Coordinates
Split
One Minus
Saturate
Power
Multiply
Vertex
Fragment
Time
Multiply
Modulo
Flipbook
Sample Texture 2D
Multiply
Split
Multiply
KnifeThrowShader
Shader Graphs
AnimationSpeed Float
TrailColor Color
Radius Float
EffectPower Float
Main Preview
Animates texture over time
Uses blurred circle as mask